

Theory of Rayleigh Fading in Wireless Autoencoders

1. What is Rayleigh Fading?

In a basic AWGN (Additive White Gaussian Noise) channel, the signal travels from the transmitter to the receiver through a direct line-of-sight path. The only degradation is the background thermal noise.

In real-world environments (like a city or inside a building), the signal bounces off buildings, cars, and trees. This phenomenon is called **Multipath Propagation**. Because the signal arrives at the receiver from multiple different paths at slightly different times, these waves interfere with each other. Sometimes they add together constructively (boosting the signal), and sometimes they add destructively (canceling the signal out, causing a “deep fade”).

When there is no strong line-of-sight path, the amplitude of the received signal fluctuates according to a **Rayleigh Distribution**. This is why it is called Rayleigh Fading.

2. Mathematical Model of the Fading Channel

In the original AWGN autoencoder, the received signal r was modeled as:

$$r = x_{norm} + n$$

Where x_{norm} is the transmitted signal and n is the Gaussian noise.

In a **Rayleigh Fading Channel**, we introduce a multiplicative fading coefficient h :

$$r = h \cdot x_{norm} + n$$

- **h (Fading Coefficient):** A random variable representing the channel's attenuation. For a real-valued baseband simulation, h is drawn from a Rayleigh distribution. We generate this by taking the magnitude of a complex Gaussian random variable:

$$h = \frac{\sqrt{X^2 + Y^2}}{\sqrt{2}}$$

- (where X and Y are standard normal random variables).
- When h is near 1, the signal passes through relatively untouched.
- When h is near 0, the signal undergoes a “deep fade”, meaning it is almost entirely destroyed before the noise n is even added.

3. The Challenge of Fading

Fading makes communication significantly harder. If the neural network transmits a point at coordinates $(1.0, 1.0)$, and the channel experiences a fade of $h = 0.2$, the received point before noise shrinks to $(0.2, 0.2)$. Because the points are now crushed much closer

to the origin (and to each other), the AWGN noise can easily push the received point across the decision boundary, causing a decoding error.

4. Perfect Channel State Information (CSI)

To combat fading, modern wireless receivers constantly estimate the channel. They do this by sending known “pilot” symbols. The receiver compares what it received to what it knows the pilot should look like, and calculates h .

In our autoencoder, we simulate **Perfect CSI at the Receiver**. This means we assume the receiver perfectly knows the value of h for every symbol.

5. How the Autoencoder Adapts (The Code Changes)

To give the neural network receiver the CSI, we explicitly pass the fading coefficient h into the decoder network.

Original AWGN Decoder:

Input dimension = `n_dims` (just the received coordinates I and Q).

```
self.decoder = nn.Sequential(
    nn.Linear(self.n_dims, self.M), ...
)
y = self.decoder(r)
```

New Rayleigh Fading Decoder:

Input dimension = `n_dims + 1` (the received coordinates I, Q, **plus** the fading value h).

```
self.decoder = nn.Sequential(
    nn.Linear(self.n_dims + 1, self.M), ...
)
# Concatenate the received signal 'r' and the fading coefficient 'h'
decoder_input = torch.cat([r, h], dim=1)
y = self.decoder(decoder_input)
```

By concatenating h , the neural network acts as a **Channel Equalizer**. It mathematically learns to scale its decision boundaries based on the current severity of the fade. If it sees $h = 0.2$, it knows the constellation has shrunk, and it will adjust its internal weights to look for points closer to the origin.

6. Geometric Constellation Effects

Because we only pass h to the *decoder* (and not the *encoder*), the transmitter doesn't know what the fading will be. It only knows that fading *will happen on average*.

Because of our strict power constraint (average power = 1), the best mathematical strategy for the transmitter remains the same: push all 16 messages as far away from each other as possible. Thus, the transmitter will still learn a **circular 16-PSK constellation**.

However, when you plot the *received* constellation, it no longer looks like a neat circle with fuzzy noise clouds. It looks like “streaks” pointing toward the origin. This is because the fading coefficient h continuously scales the circle inward and outward before the noise scatters it.

7. Performance (BLER vs E_b/N_0)

If you compare the Block Error Rate (BLER) graph of the AWGN autoencoder to the Rayleigh autoencoder, you will notice the Rayleigh curve is much flatter and requires a much higher E_b/N_0 (more power) to achieve the same error rate. This exactly mirrors theoretical communication science: fading channels severely degrade reliability compared to pure AWGN channels.